CLAVICLE

- "Collar bone"
- Medially - Sternum + 1st rib cartilage
- Laterally - Acromion process

FEATURES

- Only long bone lying horizontally
- No medullary cavity
- Membrocartilaginous ossification
- Maybe pierced through cutaneous nerve "INTERMEDIATE SUPRACLAVICULAR NERVE"
- Subcutaneous throughout its extent
- Only long bone with 2 primary centres.
- First to ossify (B/w 5th-6th wk of IUL)
  Last to ossify (25 years)

PARTS 2 ends + Shaft

ENDS

- Medial / Sternal
  - Quadrilateral
  - with clavicular notch of manubrium sterni
- Lateral / Acromial
  - Flattened
  - with acromion process margin

SHAFT

- curved
- Medial 2/3rd round & convex forward
- Lateral 1/3rd flattened & concave forward
- Inferior surface Subclavian groove in middle third

LATERAL / ACROMIAL END

- Flattened above downward
- Oval facet articulate with medial margin of acromion and form 'Acromioclavicular Joint'
- Lateral end gives attachment to fibrous capsule of acromioclavicular joint.

MEDIAL / STERNAL END

- Saddle shaped articular surface with clavicular notch of manubrium sterni to form 'Sternoclavicular Joint'
- Gives attachment to
  - Fibrous capsule
  - Articular disc
  - Interclavicular ligament

SHAFT

LATERAL 1/3rd :- Flattened, 2 surface and 2 borders

⇒ SURFACE

- **SUPERIOR**
  - Subcutaneous b/w attachment of deltoid and trapezius.
  - 1. Conoid Tubercle
    - Near posterior border at junction of lateral 1/4th and Medial 3/4th of clavicle
  - 2. Trapezoid ridge
    - Extend forward & laterally from conoid tubercle

  Attachment to 'Coracoclavicular ligament'
- **INFERIOR**
  - Origin to deltoid muscle
  - Deltoid tubercle maybe present

⇒ BORDER

- **ANTERIOR**
  - Concave forward
  - Insertion to trapezius muscle
- **POSTERIOR**
  - Convex forward
  - Origin of clavicular head of sternocleidomastoid muscle

MEDIAL 2/3rd :- Cylindrical, Has 4 surface

⇒ SURFACE

- **SUPERIOR**
  - Attachment of costoclavicular ligament to oval impression
  - Insertion of subclavius muscle into subclavian groove.
  - Attachment of clavipectoral fascia to margin of subclavian groove
  - Presence of nutrient foramen on lateral end of subclavian groove
- **INFERIOR**
  - Convex forward
  - Origin to clavicular head of pectoralis major muscle.
- **ANTERIOR**
  - Concave backward
  - Origin to sternohyoid muscle.
  - Lateral part - forms anterior boundary of cervico-axillary canal.
    - Trunks of brachial plexus
    - 3rd part of subclavian artery.
- **POSTERIOR**

SCAPULA

"Shoulder blade" . Large, flattened, triangular bone . highly mobile • Upper part of postero lateral aspect of thorax, against 2nd - 7th ribs

PARTS 4 parts
- Body
- Processes spinous, acromion, coracoid

Body 2 surface + 3 borders + 3 angles

SURFACE

COSTAL SURFACE (Subscapular fossa)
- Concave & directed medially & forward
- 3 longitudinal ridge — attachment to subscapularis muscle — arising from medial 2/3rd except near subscapular bursa.

DORSAL SURFACE
- convex
- Presents spinous process (shelf-like)
  - supraspinatous fossa (1/3rd)
  - infraspinatus fossa (2/3rd)
- Spinoglenoid notch — suprascapular nerve & vessels.
- Origin of supraspinatus and infraspinatous muscle from medial 2/3rd of supraspinatous fossa
- Origin of teres minor muscle from upper 2/3rd of dorsal surface of lateral border. Circumflex scapular artery interrupt its origin.
- Origin of teres major muscle from lower 1/3rd of dorsal surface of lateral border & inferior angle.
- Origin of latissimus dorsi muscle

BORDERS

SUPERIOR BORDER
- Shortest border
- Suprascapular notch on coracoid process — into suprascapular foramen by superior transverse ligament.
- Above the ligament ↳ suprascapular artery
  Below the ligament ↳ suprascapular nerve
- Origin of inferior belly of omohyoid near suprascapular notch

LATERAL BORDER
- Thickest border
- Presence of infraglenoid tubercle at upper end
- Origin of long head of triceps from this tubercle.

MEDIAL BORDER
- Thin and angled at root of spine
- Insertion of serratus anterior on costal surface.
- Insertion of levator scapulae on dorsal aspect.
- Insertion of rhomboideus minor on dorsal aspect
- Insertion of rhomboideus major on dorsal aspect.

ANGLES
- INFERIOR → Lies over 7th rib or 7th intercostal space.
- SUPERIOR → Junction of superior and medial borders • Lies over 2nd rib.
- LATERAL (Head of scapula)
  - → Glenoid cavity + head of humerus → 'Glenohumeral Joint' (Shoulder Joint)
  - → Glenoid labrum → margin of glenoid cavity to deepen its concavity.
  - → Capsule of shoulder joint → margin of glenoid cavity, proximal to attachment of glenoid labrum.
  - → Origin of long head of biceps brachii from supraglenoid tubercle. Origin is intracapsular.

PROCESSES

SPINOUS PROCESS :- (Spine of Scapula)

- Junction of Upper 1/3rd and lower 2/3rd.
- Upper Supraspinous fossa & Lower Infraspinous fossa.
- Has 2 surface & 3 borders.

- **SURFACE**
  - **SUPERIOR**
    - Lower boundary of supraspinous fossa
    - Origin to supraspinatus.
  - **INFERIOR**
    - Upper boundary of infraspinous fossa
    - Origin to infraspinatus.
- **BORDERS**
  - **ANTERIOR**
    - Attached to dorsal surface
  - **LATERAL**
    - Spinoglenoid notch — through which passes suprascapular nerve and vessels.
  - **POSTERIOR**
    - 'Crest of spine'.
    - Insertion of trapezius on the upper lip of crest of spine.
    - Origin of posterior fibers of deltoid from lower border of crest of spine.

ACROMION PROCESS :- (Acromion)

- Projects forward almost at right angle from lateral end of spine & overhangs glenoid cavity.
- Superior surface — subcutaneous.
- Has tip, 2 borders, 2 surface.

- **SURFACE**
  - **SUPERIOR**
    - Rough & subcutaneous
  - **INFERIOR**
    - Smooth
    - Related to 'Subacromial bursa'.
- **BORDERS**
  - **MEDIAL**
    - Continues to form upper lip of crest of spine
    - Insertion to trapezius muscle
    - Circular facet articulate with lateral end of clavicle to form 'Acromioclavicular Joint'.
  - **LATERAL**
    - Continues to form lower lip of crest of spine
    - Origin to intermediate fibers of deltoid muscle.
- Attachment of coracoacromial ligament to the tip of acromion.
- Acromial angle — junction of lateral border of acromion and lateral border of crest of spine of scapula.

CORACOID PROCESS :-

- From upper part of head of scapula
- Bent sharply to project forward and slightly laterally.
- Origin of short head of biceps brachii and coracobrachialis from tip by a common tendon.
- Insertion of pectoralis minor muscle on medial border of upper surface.
- Attachment of coracoacromial ligament to its lateral border.
- Coracoclavicular ligament
  - conoid part (rhomboid ligament) — knuckle.
  - trapezoid part (rhomboid ligament) — ridge on superior aspect b/w pectoralis minor muscle and coracoacromial ligament.
- Coracohumeral ligament — root adjacent to glenoid cavity.

HUMERUS

PARTS 2 ends + Shaft

UPPER END

1. Head : · smooth & rounded
   · forms glenohumeral joint
2. Neck
3. Greater tubercle
4. Lesser tubercle
5. Intertubercular sulcus.

SHAFT

· B/w upper & lower end
· Upper half - cylindrical
  Lower half - flattened
  - anteroposteriorly

LOWER END

1. Capitulum (lateral round convex projection)
2. Trochlea (medial pulley shaped)
3. Radial fossa (above capitulum)
4. Coronoid fossa (above trochlea)
5. Medial epicondyle
6. Lateral epicondyle
7. Olecranon fossa (deep hollow on posterior aspect above trochlea)

UPPER END

→ 1. HEAD :- · Smooth, round, 1/3rd of sphere · Covered by articular hyaline cartilage

→ 2. NECK
- → Anatomical :- · Provides attachment to capsular ligament of shoulder joint except, superiorly — for passage of tendon of long head of biceps brachii
  medially — capsule extends down for about 1 - 2 cm.
- → Morphological :- · B/w diaphysis and epiphysis · true junction of head with shaft
  · Represented by epiphyseal line
- → Surgical :- · Below greater and lesser tubercles / epiphyseal line
  · Related to axillary nerve and posterior & anterior circumflex humeral vessels.
  · One of sites of common humeral fractures.

→ 3. GREATER TUBERCLE :- · Most lateral part of proximal end
· Posterosuperior aspect — flattened facet like impressions
  - upper - supraspinatus muscle
  - middle - infraspinatus muscle
  - lower - teres minor muscle

→ 4. LESSER TUBERCLE :- · Small elevation on front of upper end (just above surgical neck)
· Subscapularis muscle.

→ 5. INTERTUBERCULAR SULCUS :- (Bicipital Groove)
· Vertical groove
· Contains
  1. long head of biceps in synovial sheath
  2. Ascending branch of anterior circumflex humeral artery.
· Lateral lip of groove - Pectoralis major muscle
  Floor of groove - Latissimus dorsi muscle
  Medial lip of groove - Teres major muscle

SHAFT

- Upper part - cylindrical, lower part - triangular
- has 3 borders and 3 surface

SURFACE

Antero-lateral
- V-shaped deltoid tuberosity
- Insertion to deltoid muscle.

Antero-medial
- Upper part — forms floor of intertubercular sulcus.
- Nutrient foramen — about its middle and close to medial border — directing downward.

Posterior
- Upper 1/3rd - oblique ridge directing downward and laterally.
  - origin to lateral head of triceps brachii.
- Below & medial to oblique ridge — radial/spiral groove
  - Radial nerve & profunda brachii vessels.
- Below spiral groove
  - origin to medial head of triceps brachii.

BORDERS

Anterior
- From lateral lip of intertubercular sulcus to anterior margin of deltoid tuberosity and ends in radial fossa

Medial
- Medial supracondylar ridge — medial intermuscular septum
- Rough strip on middle — insertion to coracobrachialis muscle
- Narrow area above medial epicondyle
  - origin to humeral head of pronator teres muscle.

Lateral
- Upper part - indistinct, lower part - lateral supracondylar ridge
- Above its middle — crossed by radial groove from behind.
- Lower part - lateral supracondylar ridge — lateral intermuscular septum.

LOWER END

- Capitulum articulate with head of radius
- Trochlea articulate with trochlear notch of ulna.
- Ulnar nerve — posterior surface of medial epicondyle.
- Anterior surface of medial epicondyle
  - common flexor origin of superficial flexors of forearm
- Anterolateral part of lateral epicondyle - common extensor origin
- Posterior surface of lateral epicondyle - Origin to anconeus muscle.
- Coronoid process of ulna — coronoid fossa (above trochlea) during full flexion
- Head of radius — radial fossa (above capitulum) during full flexion
- Olecranon process of ulna — olecranon fossa (posterior aspect above trochlea) during full extension
- Capsule of elbow joint is attached:
  Anteriorly: Above coronoid and radial fossae
  Posteriorly: Above olecranon fossa

RADIUS

- Lateral bone of forearm
- Homologous to tibia

PARTS - 2ends + Shaft

UPPER END

- Head + Neck + Radial Tuberosity
- Head : - disc shaped
  - articulate with capitulum of humerus
- Neck : - constricted
- Radial Tuberosity : - just below medial part of neck

SHAFT

- Presents lateral convexity
- Sharpest interosseous border on medial side

LOWER END

- widest part
- 5 surfaces
- Lateral surface - styloid process
- Dorsal surface - Lister's tubercle

UPPER END

- HEAD :- . Humero-radial articulation . Annular ligament
- NECK :- . Quadrate ligament
- RADIAL TUBEROSITY :- . Insertion of tendon of biceps brachii

SHAFT

- Has 3 borders & 3 surfaces.

SURFACE
- Anterior
  - Origin of Flexor policis longus — Upper 2/4th.
  - Insertion of Pronator quadratus — lower 1/4th.
  - Presence of nutrient foramen
    Nutrient artery — Anterior interosseous artery's branch.
- Posterior
  - Origin of Abductor Policis Longus (APL) — middle 1/3rd
  - Origin of Extensor policis brevis (EPB)
- Lateral
  - Insertion of Supinator — upper 1/3rd
  - Insertion of Pronator teres

BORDERS
- Anterior
  - Anterior oblique line — Origin of Flexor Digitorumsuperficialis (FDS)
- Posterior
  - Posterior oblique line
- Medial (Interosseous border)
  - Sharpest
  - Attachment of interrosseous membrane.

LOWER END

- Widest part with 5 surface

SURFACE
- Anterior
  - Palmar radio-carpal ligament of wrist joint
- Posterior
  - Dorsal tubercle of radius (Lister's tubercle) — for tendon of extensor policis longus.
  - Lateral to groove - ECRL & ECRB
    Medial to groove - Extensor policis longus — passes tendons of extensor digitorum and extensor indicis.
- Medial
  - Articular disc of inferior radio-Ulnar joint
- Lateral
  - Tendons of Abductor pollicis longus & Extensor pollicis brevis.
  - Insertion of brachioradialis
- Inferior
  - Lateral Δr area — scaphoid
    Medial □r area — lateral part of lunate

ULNA

- Medial bone of forearm
- Homologous to fibula.

PARTS - 2 ends + Shaft

UPPER END

- Trochlear notch b/w
  - Above : Olecranon process
  - Below : Coronoid process

SHAFT

- Lateral border (interosseous border) is sharp crest like.

LOWER END

- Has head & styloid process

UPPER END - Has 2 process & 2 notches.

- PROCESS → OLECRANON - Has 5 surface
  - SURFACE → UPPER - Insertion to triceps brachii - Capsular ligament of elbow jt.
  - SURFACE → ANTERIOR - Upper part of trochlear notch.
  - SURFACE → POSTERIOR - Subcutaneous olecranon bursa
  - SURFACE → MEDIAL - Attachments to :-
    - Origin of Flexor carpi ulnaris (FCU)
    - Posterior & Oblique bands — Ulnar collateral ligament. (UCL)
  - SURFACE → LATERAL
- PROCESS → CORONOID - Has 4 surface
  - SURFACE → SUPERIOR - Lower part of trochlear notch
  - SURFACE → ANTERIOR - Ulnar tuberosity - Insertion of Brachialis muscle
    - Sublime tubercle :-
      - Ant band of UCL
      - Oblique band of UCL
      - Humero ulnar head of FDS
      - Pronator teres
      - Flexor policis longus
  - SURFACE → MEDIAL - Origin of Flexor digitorum Profundus.
  - SURFACE → LATERAL - Radial notch
    - Supinator fossa — Accommodates Radial tuberosity (during supination & pronation)
    - Origin to Supinator muscle
- NOTCHES → TROCHLEAR - Semilunar - With trochlea of humerus
  - Attachments to capsule of elbow joint.
- NOTCHES → RADIAL - Superior radio ulnar joint.

SHAFT - 3 surface + 3 borders

- SURFACE → ANTERIOR
  - Origin of Flexor digitorum profundus (Upper 3/4th)
  - Origin of Pronator quadratus (Lower 1/4th)
  - Presence of nutrient foramen
- SURFACE → MEDIAL - Origin of Flexor digitorum profundus (Upper 2/3rd)
- SURFACE → POSTERIOR
  - Insertion of Anconeus (above oblique line)
  - Below oblique line — larger medial, smaller lateral
  - From proximal to distal at lateral part — attachments to :
    1. Abductor policis longus
    2. Extensor policis longus
    3. Extensor indicis

BORDERS
- LATERAL (interosseous)
  - Sharpest
  - Attachment to interosseous membrane
- ANTERIOR
  - Origin to Flexor digitorum profundus
- POSTERIOR
  - Subcutaneous
  - Attachment by common aponeurosis
    1. Flexor digitorum profundus
    2. Flexor carpi ulnaris
    3. Extensor carpi ulnaris

LOWER END
- Head + Styloid process

- HEAD
  - Ulnar notch of radius — Inferior radio ulnar joint
- STYLOID PROCESS
  - Tip — medial collateral ligament of wrist joint
  - Apex — depression b/w head & base of process
  - Groove — tendon of extensor carpi ulnaris

SACRUM

GENERAL FEATURES

1. BASE 2. APEX 3. SURFACES 4. SACRAL CANAL

BASE

- Upper surface articulates with L5 vertebra → Lumbosacral Joint
- Vertebral foramen → sacral canal
- Structures passing Ala of sacrum from Lateral to medial
  - a. Obturator nerve
  - b. Iliolumbar artery
  - c. Lumbosacral trunk
  - d. Sympathetic chain
- Origin to Iliacus muscle
- Attachment to lumbosacral ligament

APEX

- Oval facet → with coccyx → Sacrococcygeal joint

SURFACES

→ PELVIC
- Median sacral vessels
- Sympathetic trunks.
- Origin of piriformis muscle
- Pelvic sacral foramina

→ DORSAL
- Transmits upper 4 sacral spinal nerves
- Origin of erector spinae muscles. and multifidus.

→ LATERAL
- Sacroiliac Joint → Sacroiliac ligament.

COCCYX

- Represents tail ('tail bone')

GENERAL FEATURES

1. BASE
2. APEX OR TIP
3. PELVIC SURFACE
4. DORSAL SURFACE
5. LATERAL MARGIN

BASE
- Cartilaginous Sacrococcygeal joint
- 5th Sacral nerves

APEX
- Attachment to sphincter ani externus and anococcygeal ligaments.

PELVIC SURFACE
- Attachment to coccygeus muscle and levator ani
- Presence of ganglion impar.

DORSAL SURFACE
- Origin of Gluteus maximus – each side
- Origin of Sphincter ani externae – tip
- Attachment to filum terminale

LATERAL MARGINS
- Attachment to sacrotuberous and sacrospinous ligaments.

≫ FEMUR ≪

· Longest & strongest

PARTS

UPPER END
· Head + Neck + Greater and Lesser trochanter

SHAFT
· Narrowest in middle

LOWER END
· Medial + Lateral condyles
· Intercondylar fossa

UPPER END

→ HEAD
· Acetabulam of hip bone — hip joint
· Fovea — ligamentum teres femoris.

→ NECK · 2 borders and 2 surface

SURFACE → ANTERIOR
· Flat + no. of oblique bony ridges
· intertrochanteric line
· Completely intracapsular

★ INTERTROCHANTERIC LINE
· Spiral line
· Attachment to :
1. Capsule of hip joint
2. Iliofemoral ligament
3. Upper end : Vastus lateralis
Lower end : Vastus medialis

→ POSTERIOR
· intertrochanteric crest
· Medial half — intracapsular
· Tendon of obturator externus.

★ INTERTROCHANTERIC CREST
· Quadrate tubercle — Quadratus femoris

BORDER → UPPER · Greater trochanter
→ LOWER · Lesser trochanter.

→ GREATER TROCHANTER · 1 border & 3 surface

BORDER → UPPER · Piriformis

SURFACE → ANTERIOR
· Gluteus minimus - Lateral
· Trochanteric bursa of Gluteus minimus — Medial

→ MEDIAL
· Trochanteric fossa — obturator externus.
· Insertion of obturator internus + Gemellus superior and Gemellus inferior.

→ LATERAL
· Ridge — Gluteus medius muscle
· Δr areas — Anterior - Gluteus medius
Posterior - Gluteus maximus

→ LESSER TROCHANTER
· Insertion to Psoas major
· Attachment of Iliacus.

SHAFT

- It has 5 surface & 3 borders

SURFACE → ANTERIOR
→ MEDIAL
→ LATERAL
→ UPPER POSTERIOR
→ LOWER POSTERIOR (POPLITEAL SURFACE)

BORDERS → MEDIAL
→ LATERAL
→ POSTERIOR

LINEA ASPERA (Posterior border)

- Has inner lip + outer lip + intermediate area

UPPER END :-
⇒ Inner Medial lip → spiral line
⇒ Outer lateral lip → Gluteal tuberosity
⇒ Posterior surface

LOWER END :-
⇒ Inner Medial lip → Medial supracondylar line → Adductor tubercle
⇒ Outer lateral lip → Lateral supracondylar line
⇒ Popliteal surface

ATTACHMENTS → INTERMUSCULAR SEPTA
- Medial intermuscular septum – Medial lip
- Lateral intermuscular septum – Lateral lip

→ MUSCULAR
- Vastus intermedius
- Articularis genu
- Vastus lateralis
- Vastus medialis
- Gluteus maximus
- Adductor longus
- Pectineus
- Adductor brevis
- Adductor magnus
- Gastrocnemius
- Plantaris

LOWER END

MEDIAL CONDYLE
- Attachment to upper end of medial collateral ligament
- Adductor tubercle - ischial head of adductor magnus

INTERCONDYLAR FOSSA
- Medial wall – upper end of posterior cruciate ligament
- Lateral wall – upper end of anterior cruciate ligament

[LAMP]

LATERAL CONDYLE
- Attachment to fibular collateral ligament of knee jt.
- Origin of lateral head of gastrocnemius.
- Attachment to popliteus

TIBIA

. Also known as 'Shinbone' . Second largest bone . Homologous to Radius

PARTS

UPPER END
. Medial & lateral condyles + tibial tuberosity

SHAFT
. 3 borders & 3 surface

LOWER END
. Medial malleolus

UPPER END . Has 2 condyles + intercondylar area + tibial tuberosity.

→ MEDIAL CONDYLE . 4 surface

- SURFACE → SUPERIOR
  - . Articulate with femoral condyle
  - . Medial meniscus (fibrocartilaginous plate)
- → POSTERIOR
  - . Insertion of tendon of semimembranosus.
- → ANTERIOR
- → MEDIAL
  - . Vascular foramina
  - . Attachment to medial patellar retinaculum.

→ LATERAL CONDYLE . 4 surface

- SURFACE → SUPERIOR
  - . Articulate with lateral femoral condyle
  - . Lateral meniscus
- → POSTERIOR
  - . Articular facet for head of fibula
  - . tendon of popliteus.
- → ANTERIOR
  - . Flat or facet – iliotibial tract
- → LATERAL
  - . Along with anterior surface – attachment to lateral patellar retinaculum.

→ INTERCONDYLAR AREA
- . Intercondylar eminence → intercondylar tubercles
- . Attachments to
  1. Anterior horn of Medial meniscus
  2. Anterior cruciate ligament
  3. Anterior horn of Lateral meniscus
  4. Posterior horn of Medial meniscus
  5. Posterior horn of Lateral meniscus
  6. Posterior Cruciate ligament

→ TIBIAL TUBEROSITY
- . Upper smooth – ligamentum patellae
  Lower rough – subcutaneous infrapatellar bursa
- . Deep infrapatellar bursa.

SHAFT . has 3 borders & 3 surfaces

- BORDERS → ANTERIOR
  - . Sharpest, subcutaneous, sinous
  - . Tibial tuberosity → Anterior border of medial malleolus.
- → MEDIAL
  - . Medial condyle → Posterior border of medial malleolus
- → LATERAL (Interosseous border)
  - . Lateral condyle → Anterior border of fibular notch.
  - . Attachment to interosseous membrane uniting tibia & fibula
  - . Attachment of interosseous inferior tibiofibular ligament.

SURFACE → MEDIAL
- Almost entirely subcutaneous
- Upper part — insertion of
  - S - Sartorius
  - G - Gracilis
  - S - Semitendinous

→ LATERAL
- Upper 2/3 — origin to tibialis anterior
- Lower 1/3 - medial → lateral
  1. Tibialis anterior
  2. Extensor hallucis longus
  3. Anterior tibial artery
  4. Anterior tibial nerve / Deep peroneal nerve
  5. Extensor digitorum longus
  6. Peroneus tertius.

→ POSTERIOR
- Soleal line — from above downward
  1. Fascia covering popliteus
  2. Fascia covering soleus
  3. Soleus
  4. Deep transverse fascia
- Δr area above soleal line — insertion to popliteus
- Nutrient foramen — Nutrient artery branch of posterior tibial artery — largest nutrient artery of body.
- Flexor digitorum longus - medial line below soleal line
- Tibialis posterior — lateral area below soleal line.
- Lower 1/4th - medial to lateral
  1. Tibialis posterior
  2. Flexor digitorum longus
  3. Posterior tibial artery
  4. Tibial nerve
  5. Flexor hallucis longus

LOWER END - 5 surface & Medial Malleolus.

SURFACES
→ ANTERIOR
→ POSTERIOR - Groove for tendon of tibialis posterior
→ MEDIAL
→ LATERAL - Fibular notch — anterior and posterior tibiofibular ligaments.
upper part — interosseous tibiofibular ligament
→ INFERIOR - Articulate with body of talus.

MEDIAL MALLEOLUS
- Attachment to deltoid ligament.
- Lateral surface — medial surface of body of talus
- Posterior surface — vertical groove for tendon of tibialis posterior.
- Medial margin — flexor retinaculum.

FIBULA

- Homologous with ulna
- Pin-like bone

PARTS

UPPER END
- styloid process

SHAFT

LOWER END
- Malleolar fossa

UPPER END

→ HEAD
- Oval or circular articular facet
- Styloid process → fibular collateral ligament
- Sloping surface → insertion of biceps femoris

→ NECK
- Posterolateral → Common peroneal nerve
- Medial → Anterior tibial artery.

LOWER END

- 4 surface

SURFACE
- → ANTERIOR
  - Anterior talofibular ligament
  - Calcaneofibular ligament
- → POSTERIOR
  - tendons of peroneus brevis & longus.
- → MEDIAL
  - Malleolar fossa
  - Upper → posterior tibiofibular ligament
  - Lower → posterior talofibular ligament
- → LATERAL
  - Subcutaneous.

SHAFT
- 3 borders & 3 surface.

SURFACE
- → MEDIAL (Extensor)
  - Origin to extensor digitorum longus, extensor hallucis longus, peroneus tertius.
- → LATERAL (Peroneal)
  - Origin of Peroneus longus and brevis
- → POSTERIOR (Flexor)
  - Origin of tibialis posterior from medial concave part
  - Origin of soleus and flexor hallucis longus from lateral flattened part
  - Peroneal artery descends along medial crest
  - Presence of nutrient artery.

BORDERS
- → ANTERIOR
  - Attachment of superior extensor retinaculum and superior peroneal retinaculum
- → POSTERIOR
  - Attachment of posterior intermuscular septum.
- → MEDIAL (Interosseous)

HIP BONE

- Also known as 'Coxae', 'Innominatum'
- Fusion of Ilium, Ischium & Pubis

ILIUM

- 2 ends + 3 borders + 3 surfaces.

→ **ENDS** → Upper (Iliac crest)

Anterior superior iliac spine → Above : Inguinal ligament
Below : Sartorius

Posterior superior iliac spine → Sacrotuberous ligament

A. VENTRAL SEGMENT — Outer lip + intermediate area + Inner lip

1. Outer lip – tubercle of iliac crest
   – Attachments to
   · origin of tensor fascia latae
   · insertion of external oblique muscle
   · origin of latissimus dorsi
2. Intermediate area · Origin of internal oblique muscles.
3. Inner lip · Anterior 2/3rd : Transversus abdominis muscle
   · Posterior 1/3rd : Quadratus lumborum

B. DORSAL SEGMENT

1. Outer sloping surface : Gluteus maximus muscle
2. Inner sloping surface : Erector spinae muscle

→ Lower

BORDERS → Anterior · Anterior inferior iliac spine
Upper : straight head of rectus femoris
Lower : iliofemoral ligament

→ Posterior · Posterior inferior iliac spine
Greater sciatic notch → foramina
Ischial spine
Lesser sciatic notch → foramina
· Structures passing from dorsal aspect of ischial spine (Lateral → medial)
– Pudendal nerve
– Internal pudendal vessels
– Nerve to obturator internus.

→ Medial · Arcuate line

SURFACE → Iliac fossa · Origin of iliacus

→ Gluteal · 4 areas by 3 gluteal lines.
· Origin to 4 muscles

Gluteus maximus behind posterior gluteal line
Gluteus medius b/w anterior and posterior gluteal line
Gluteus minimus b/w anterior and inferior gluteal line
Rectus femoris b/w inferior gluteal line and upper margin of acetabulum.

→ SACROPELVIC
- A. SACRAL PART
  1. Iliac tuberosity – attachments to ligaments
     a. iliolumbar
     b. dorsal sacroiliac
     c. interosseous sacroiliac
  2. Auricular surface – sacroiliac joint
- B. PELVIC PART – Pre auricular sulcus
  – Origin to obturator internus.

ISCHIUM

Body – 2 ends + 3 borders + 3 surface & Ramus

→ BODY

ENDS → Upper
Lower

- Ischial tuberosity
  - Upper quadrilateral area → upper lateral / lower medial
  - Lower triangular area → outer / inner
- Origin to 4 muscles.
  1. Semimembranosus
  2. Semitendinosus
  3. Semitendinosus + long head of biceps femoris
  4. Adductor magnus
- Medial margin → lower end of sacrotuberous ligament.

BORDERS →
- ANTERIOR – Attachment to obturator membrane
- POSTERIOR – Ischial spine → attachments to
  - sacrospinous ligament
  - levator ani
  - coccygeus muscle
  - Lesser sciatic notch
    - upper end : Gemellus superior
    - lower end : Gemellus inferior
- LATERAL – tendon of obturator externus.

SURFACE →
- Femoral – Obturator externus – along obturator foramen
  Quadratus femoris – along lateral border of ischial tuberosity.
- Dorsal – Tendon of obturator internus + 2 gemelli — traverses the groove
- Pelvic – Origin to obturator internus.

→ RAMUS

Conjoint ischiopubic ramus – 2 borders + 2 surface

BORDERS → Upper – Attachment to Obturator membrane
Lower

SURFACE →
- Outer – Linear attachment to 4 muscles
  1. Obturator externus
  2. Adductor magnus
  3. Adductor brevis
  4. Gracilis
- Inner – 2 ridges + 3 areas
  - Ridges → Upper – Attachment to superior fascia of urogenital diaphragm
    Lower – Attachment to perineal membrane & falciform process of sacrotuberous ligament.

- Area → Upper — origin to obturator internus
  - Middle — Attachment to sphincter urethrae & deep transversus perinei muscles.
  - Lower — Attachments to crus of penis, clitoris, ischiocavernosus & superficial transversus perinei muscles.

PUBIS

- 'Pubic bone / OS Pubis'.
- Body + superior ramus + inferior ramus.

→ BODY

- Pubic crest + Pubic tubercle + 3 surface.

- PUBIC CREST & PUBIC TUBERCLE
- SURFACE
  - Anterior
    - Attachments to 4 muscles
      1. Adductor longus
      2. Gracilis
      3. Adductor brevis
      4. Obturator externus
  - Posterior (Pelvic) — Related to urinary bladder.
  - Medial (Symphyseal) — Symphysis pubis – 2° cartilaginous joint.

→ SUPERIOR RAMUS

- 3 borders + 3 surfaces.

- BORDERS
- → Pectineal line (Posterior / superior)
  - 'Pecten pubis'
  - Attachments to
    - Medial
      - Lacunar ligament
      - inguinal ligament
      - conjoint tendon
      - Fascia transversalis
    - Lateral
      - Pectineal ligament (Ligament of cooper)
      - Pectineus muscle
      - pectineus fascia
      - Psoas minor (if present)
- → Obturator crest (Anterior)
- → Inferior — Attachment to obturator membrane leaving a gap for obturator canal.

SURFACE
- Pectineal — Origin of pectineus muscle.
- Pelvic — Featureless & covered by peritoneum.
- Obturator — Obturator nerve & vessels traverses the groove.

→ INFERIOR RAMUS

- form conjoint ischiopubic ramus.

OBTURATOR FORAMEN

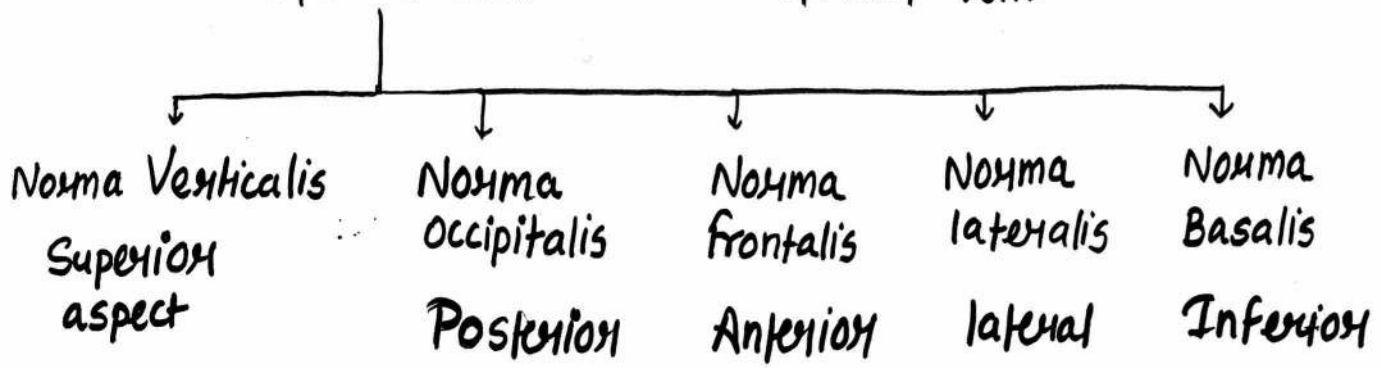
- Attachment of obturator membrane except the groove to convert it into obturator canal
- Obturator canal transmits obturator nerves & vessels.

ACETABULUM

- Shallow vinegar cup
- Acetabular notch → acetabular foramen due to transverse acetabular ligament
- Articular – Lunate surface
  Non articular area – Acetabular fossa
- Acetabular labrum (fibrocartilaginous rim) → margin of acetabulum
- Acetabular fossa – filled with pad of fat
  Lunate surface – covered by hyaline cartilage

Study of Skull done from Outside or inside

- Exterior Veiw
- Interior Veiw



Norma Verticalis

→ oval
→ Wider Posteriorly than anteriorly.
→ Bones 1. Frontal Bone → Squamous Part
2. Parietal Bone → Paired
3. Occipital Bone → Squamous Part

→ Sutures 1. Coronal Suture → B/W Fronal and 2 Parietal Bones
2. Sagittal Suture → B/W 2 Parietal Bones
3. lamdoid Suture → B/W occipital and 2 Parietal Bones

★ Metopic Suture in 3 % - 8 % cases Present in Median Plane

→ Other Features 1. Bregma → Point where Sagittal and Coronal Sutures meet
2. Parietal Eminence / Tuber → Area of Maximum Convexity in Parietal Bone
3. Vertex → highest Point on Skull
4. Parietal Foramen → • Present near Sagittal Suture
• transmits emissary Vein
5. Obelion → Point B/W 2 Parietal foramen.

Norma Occipitalis

Convex Upwards and sides and Flattened Below

Bones - 1. Parietal Bones - Posterion Portions of Parietal Bone Superiorly One on each side

2. Occipital Bone - Upper Part of Occipital Bone Present inferiorly.

3. Mastoid Part of temporal Bone - Present inferolaterally on each side

Sutures - • lambdoid suture • Occipitomastoid Suture
• Parietomastoid Suture

Other Features • lamda
• External Occipital protubrence
• Superion nuchal lines
• inferion nuchal lines
• highest nuchal linus (some Cases)
• Mastoid Foramen - Present on Posterion aspect of Mastoid Process near Occipito-mastoid suture
- transmits → Emissary Vein

Norma Frontalis

Bones - 1. Frontal Bone
2. Nasal Bone ┐
3. Maxillae ┘ → Right and left
4. Zygomatic Bone (Right and left)
5. Mandible

Sutures - 1. Frontonasal
2. Internasal
3. Frontomaxillary
4. Zygomaticomaxillary

anterion aspect of Skull Presents

1. Frontal Region formed By Squamous Part of frontal Bone
2. Orbital openings
3. Prominences of the cheek formed By Zygomatic Bones
4. Bony External nose and Anterion Nasal Aperture.
5. Upper and lower Jaw Bearing teeth.

Foramen :- 1. Supraorbital Foramen / Notch
- Supraorbital nerve and Vessels
- Frontal diploic vein.

2. Superior Orbital Fissure
- lacrimal artery and nerve
- Frontal nerve
- Trochlear nerve
- ABducent nerve
- Nasociliary nerve
- Superior and inferior division of Occumulator nerve
- Sympathetic Roots of Ciliary ganglions
- Superior opthalmic Vein
- Reccurent meningeal Branch of lacrimal artery

3. Inferior orbital fissure
- Maxillary nerve
- Infraorbital Vessels
- zygomatic nerve
- Vein Connecting inferior opthalmic Vein with Pterygoid Venous Plexus.

4. Infraorbital Foramen
- Infra Orbital nerve and artery → Continuation of maxillary artery and vein.
- Infraorbital Vein

5. Zygomatico-Orbital foramen
- zygomatic Branch of maxillary nerve.

6. Mental Foramen.
- Mental nerve and Vessels.

Other features -
- Glabella - Median elevation above the nasion and B/W Superciliary arches
- Nasion - Median Point at the root of neck
- Anterior nasal Spine - Bony Projection in Median plane
- Symphysis Menti - Median Ridge Joining 2 halves of Mandible
- Mental Protuberance - Δ elevation at lower end of Symphysis Men
- Mental Point - Middle point at the Base of mandible
- Frontal Prominence - Rounded elevation above Superciliary arch

Foramina and Structure Passing through :-

1. Incisive foramina - greater palatine vessel
   - nasopalatine nerve
2. Greater palatine formen - greater palatine nerves and vessels
3. Lesser palatine foramina - Middle and Posterior palatine nerve
4. Formen ovale - Mandibular nerve
   Accessory middle meningeal artery
   lesser Petrosal nerve
   Emissary vein
5. Foramen Rotendum - Maxillary nerve
   - Emissary Vein.
6. Foramen Spinosum - Middle meningeal artery
   Middle meningeal vein
   Nerve Spinous
7. foramen lacerum → Meningeal → Ascending Pharyngeal Artery Branch
   → Emissary Vein
   → internal carotid artery
   → Greater Petrosal nerve
8. Carotid Canal → Internal carotid artery and vein
   → Emissary Vein.
9. Jugular foramen → Anterior Part - Inferior Petrosal Sinus
   → Middle part - 9th, 10th, 11th Cranial nerve
   → Posterior Part - Sigmoid Sinus

10. Stylomastoid Foramen → 7th Cranial nerve
                          → Stylomastoid artery

11. Foramen Magnum

Anterior Compartment
- Apical ligament of Dens
- Upper Band of Cruciform ligament of atlas
- Membrana tectoria

Posterior Compartment
- Medulla oblangata
- Two posterior spinal artery
- 1x anterior spinal artery
- 2x Vertebral arteries
- Sympathetic plexus
- Spinal Root of two accessory nerve

12. Hypoglossal Canal → 12th Cranial nerve
                       → Meningeal Branch of ascending Pharyngeal artery.

Norma lateralis

Bones - • Frontal • temporal • Mandible
• Parietal • Sphenoid • Maxilla
• Occipital • zygomatic • nasal

Suture - • Coronal Suture
• Parietosquamosal Suture - B/w Parietal and Squamous Part of temporal Bone
• lamdoid Suture

Other features - 1. Temporal line - Commences at the frontal Process of zygomatic Bone arches Upwards and Backwards across the Parietal Bone where it Splits in Superior and inferior lines

2. Zygomatic arch → horizantal Bar of Bone form by temporal Process of zygomatic Bone and zygomatic Process of temporal Bone.
3. External auditory Meatus
4. Suprameatal triangle - Small depression posteriosuperior to the External auditory meatus
5. Mastoid Process of temporal Bone
6. Asterion - Meeting Point of Parietomastoid, occipitomastoid and lambdoid Suture.
7. Styloid Process of temporal Bone
8. Temporal fosa and infratemporal Fosa
9. Pterion - Region in anterior Part of temporal fosa where 4 bones meet to form H Shaped Structure
   1. frontal 3. Squamous temporal
   2. Parietal 4. Greater Wing of Sphenoid.

Foramen and fissures of Norma lateralis

1. Zygomaticotemporal foramen - Zygomaticotemporal artery and nerve passes
2. Sphenopalatine foramen - Nasopalatine nerves
   Sphenopalatine Vessels
3. Pterygomaxillary fissure - 3rd part of Maxillary artery
   Maxillary Nerve
4. Greater Palatine Canal - Palatine nerves and vessels.

Interior Surface of Skull

- divided By Sphenoidal ridge (in front)
- Petrous Ridge Behind

Cranial Fosa → anterior, Middle, Posterior

- **Anterior Cranial fosa**
  - Frontal lobes of Cerebral hemispheres
  - Floor - Ethmoid, Frontal and Sphenoid
  - demarcated from middle Fosa By
    - Posterior free Border of lesser Wing of Sphenoid on each side
    - anterior Border of Sulcus chiasmaticus in median Region

Features → in median Region

- Frontal Crest → Vertical Crest on inner aspect of frontal Bone
- Foramen Caecum → in B/w frontal Crest and Crista galli
- Crista galli → Bony Crest formed By Perpendicular plate of ethmoid.
- Fugum Sphenoidale → Superior Surface of anterior part of Body of Sphenoid
- on each side of Crista galli lies the sieve-like Cribriform plate of ethmoid

→ lateral Region - on either side - Orbital plate of frontal Bone

Foramen → 1. Foramen Caecum - Emissary Vein
2. Cribriform Foramina - Olfactory nerves Surrounded By leptomeninges
3. Anterior Ethmoidal foramen - Anterior ethmoidal nerve and Vessels

Middle Cranial Fosa

→ divided from anterior cranial fosa By - Superior Border of Petrous temporal Bone on each side

- Dorsum Sellae in median Region

→ Floor (Butterfly shape) → Median Part → formed By Bod of Sphenoid
→ lateral Part

→ Features
- Sulcus chiasmaticus
- Sella Turcica

formed By
- Greater wing of Sphenoid
- anterior Surface of Petrous temporal Bone
- Squamous part of temporal Bone

Features
- Trigeminal impression
- Hiatus and groove for greater and lesser petrosal nerve
- Arcuate eminence
- Tegmen tympani

→ Foramen

Optic Foramen -
- optic nerve
- Opthalmic artery
- Sympathetic plexus around opthalmic artery

Superior orbital fissure
divided By Common tendinous Ring of Zinn

Within Common tendinous Ring
- Superior and inferior division of Oculomotor nerve
- Nasociliary nerve
- Abducent nerve
- Sympathetic Root of ciliary ganglion

through parts above and lateral of Ring
- Trochlear nerve
- Frontal nerve
- lacrimal nerve
- lacrimal artery
- Superior Opthalmic Vein
- Recurrent meningeal Branch of lacrimal artery

through Below and Medial of Ring

Foramen Rotundum
- Maxillary nerve
- Emissary Vein

Foramen ovale
- Mandibular nerve
- Accessory Middle Meningeal artery
- lesser Petrosal nerve
- Emissary Vein

Foramen Spinosum
- Middle Meningeal artery
- Nerve Spinosus

Foramen lacerum
- Emissary Vein
- Internal carotid Art.
- Meningeal Branch of ascending Pharyngeal art.

Posterior Cranial Fosa

Bones - Sphenoid, temporal, parietal and occipital

Features - Median area - • clivus - Sloping surface in front of foramen magnum formed by Posterior Part of Body of sphenoid with Basilar part of occipital Bone.

- Squamous Part of Occipital Bone
  - • foramen Magnum
- Internal occipital crest - Vertical Bony Ridge from occipital Protuberance to foramen magnum
- Internal occipital Protuberance
- Groove for transverse and sigmoid sinus
- Cerebella fosa
- Fosa for occipital lobes

- Posterior Surface of Ptrous Bone
- Internal auditory Meatus
- Subarcuate Fosa - Depression lateral and superior to Internal auditory Meatus
- Aqueduct of Vestibule - slit behind Internal auditory Meatus.
- Cochlear Canaliculus - notch on lower Border of posterior Surface of PEtrous Bone.

→ Foramina and Canals

- Internal auditory Meatus

1. facial nerve 2. Vestibocochlear nerve
3. Nervus intermedius 4. Internal auditory vessels.

- Jugular foramen

Anterior Part
- Inferior Petrosal Sinus

Middle
- 9th
- 10th
- 11th

} Cranial nerve

10th and 11th in common sheath of

Posterior
- Sigmoid Sinus

Hypoglossal Canal
- hypoglossal nerve
- Meningeal Branch of ascending pharyngeal artery.

Posterior Condylar Canal
- Emissary Vein
- Meningial Branch of Occipital artery

Foramen magnum
| divided By alar ligament

Anterior Compartment
- Apical ligament of Dens
- Upper Band of Cruciform ligament of atlas.
- Membrana tectoria

Posterior Compartment
- Medulla oblangata
- Two posterior Spinal arteries (R and L)
- Two Vertebral arteries (R and L)
- Spinal Root of 2 accessory nerves (R and L)
- Sympathetic Plexus around Vertebral arteries
- Communicating Veins B/W internal Vertebral Venous plexus and Basilar Venous plexus.

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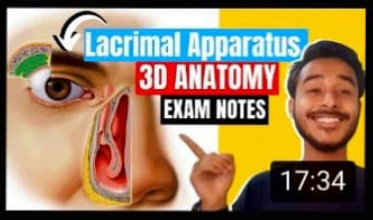
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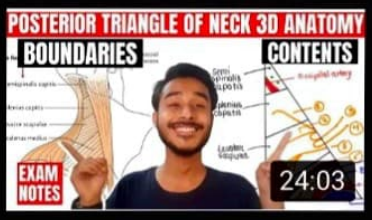
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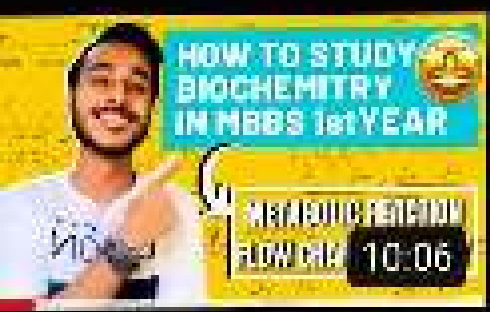
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